

★ WEEK 3 · PART 3 · PRACTICE

ARMA Forecasting, Worked by Hand

Watch the Forecast Fade to the Mean, and the Band Fan Out

Point forecasts, widening intervals, and the benchmark

Part 3's payoff, with the arithmetic shown. We turn a fitted AR(1) forward, watch its point forecast decay geometrically to the mean, see the prediction interval widen to the unconditional level - and confront the humbling truth about forecasting returns.

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PART 3 · APPLICATIONS

How Far Ahead Is Useful

Case A: iterate the fitted AR(1) forward and watch the point forecast decay to the mean as the past stops helping.
Case B: the forecast error variance grows, widening the band to the unconditional level - and a forecast is only good if it beats the naive benchmark. Then a Claude Code lab.

What Part 3 Told Us

Now we apply it - with numbers

1

Forecasts decay to the mean

A stationary model's forecast reverts to its long-run mean. The farther out you look, the less the last value helps.

2

Intervals must widen honestly

Uncertainty grows with the horizon toward the unconditional variance. A point forecast without a band is dangerously incomplete.

3

You must beat the naive benchmark

A forecast only counts if it beats the obvious one. For returns, with little structure, that bar is humbly hard to clear.

Everything here is a calculator exercise: one decaying forecast, one widening band, one benchmark.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a fitted AR(1) and its last value.
- Iterate the forecast, or grow the variance.
- Then read the horizon.

What you should watch

- How the forecast decays to the mean.
- How the band widens to the unconditional level.
- How hard it is to beat naive on returns.

The Forecast Decays

The past stops helping

With a fitted AR(1), forecasting is just iterating the model forward. The last value starts above the mean, but each step multiplies its deviation by the coefficient, so the forecast decays geometrically back to the long-run mean. We compute the point forecast at several horizons and watch the memory fade

The fitted model

- Coefficient 0.6, mean 1.0.
- The last observed value is 3.0.
- Two full points above the mean.

What we'll do by hand

- Iterate the forecast forward.
- Watch the deviation shrink by 0.6 each step.
- See it decay toward the mean.

The ARMA Forecast

Turn the fitted model forward

$$\hat{y}_{T+h} = \mu + \phi^h(y_T - \mu) \quad (\text{iterate the fitted model forward; the deviation shrinks by } \phi \text{ each step})$$

How it iterates

- The forecast is the mean plus a decayed deviation.
- Each step multiplies the gap by the coefficient.
- Future shocks are unknown, so set to zero.

The consequence

- The deviation from the mean shrinks geometrically.
- Distant forecasts carry little of the last value.
- The model forgets, at rate ϕ .

The Point Forecasts

Fading toward the mean

$$1.0 + 2(0.6)^h : \quad \mathbf{2.20} \ (h=1), \ \mathbf{1.72} \ (h=2), \ \mathbf{1.43} \ (h=3) \rightarrow \mu=1.0$$

The arithmetic

- One step: 1.0 plus 2 times 0.6, or 2.2.
- Two steps: 1.0 plus 2 times 0.36, or 1.72.
- Three steps: 1.43, and falling.

The trajectory

- Each forecast is closer to the mean.
- The initial gap of 2.0 keeps shrinking.
- By ten steps it's essentially at 1.0.

Decay to the Mean

When the past stops mattering

$$\text{as } h \rightarrow \infty : \phi^h \rightarrow 0 \Rightarrow \hat{y}_{T+h} \rightarrow \mu \quad (\text{the past stops helping})$$

The limit

- As the horizon grows, ϕ -to-the- h vanishes.
- So the forecast converges to the mean.
- The last value's influence disappears.

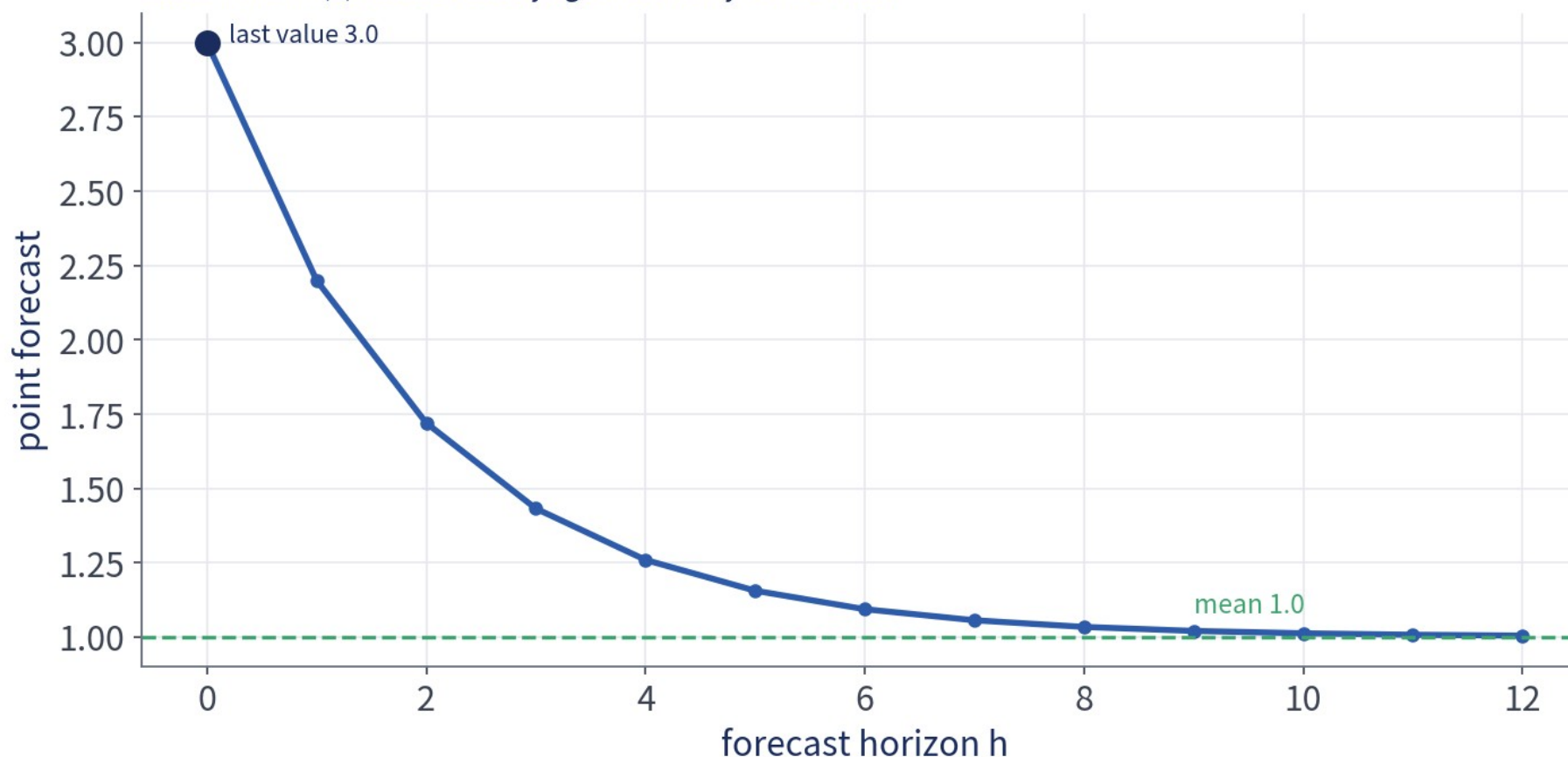
The lesson

- There's a horizon past which you learn nothing.
- Beyond it, the best guess is just the mean.
- Know where your forecast stops speaking.

The Memory Fades

Starting from a last value of 3.0, two points above the mean, the forecast decays smoothly back toward the dashed mean of 1.0. Each step erases 40% of the remaining gap. Within about ten steps the forecast is indistinguishable from the mean – the point where the past has nothing left to say

Case A: the AR(1) forecast decays geometrically to the mean



The Information Horizon

How far the model can see

What decay implies

- Short horizons carry real information.
- Long horizons collapse to the mean.
- The coefficient sets how fast.

How to use it

- Trade on the near-term forecast, not the far one.
- A high coefficient means a longer useful horizon.
- Past that, the mean is your best guess.

Honest Intervals, Hard Benchmark

The band around the bar you must have

A point forecast alone is dangerously incomplete. As the horizon grows, forecast errors accumulate, so the prediction interval widens - toward the unconditional variance. We compute that widening band, then confront the benchmark: a forecast is only useful if it beats the naive one, a bar that returns, with their thin

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The widening band

- Error variance grows with the horizon.
- It approaches the unconditional variance.
- So the band fans out, then levels off.

What we'll do by hand

- Compute the error variance at each horizon.
- Turn it into a widening interval.
- Then weigh the benchmark.

The Forecast Error Variance

Uncertainty that accumulates

$$\text{Var}(e_h) = \sigma^2 \frac{1-\phi^{2h}}{1-\phi^2} \nearrow \frac{\sigma^2}{1-\phi^2} = 1.5625 \quad (\text{the unconditional variance})$$

How it grows

- Each step adds another shock's variance.
- The sum is a geometric series.
- It climbs toward a finite limit.

The ceiling

- The limit is the unconditional variance.
- Here, 1.5625.
- Beyond it, uncertainty can't grow.

The Widening Band

From one step to the limit

horizon	error variance	std dev	95% band
$h = 1$	1.00	1.00	+/- 1.96
$h = 2$	1.36	1.17	+/- 2.29
$h = 3$	1.49	1.22	+/- 2.39
$h = \infty$	1.5625	1.25	+/- 2.45

The error variance climbs from 1.0 at one step toward the unconditional 1.5625, so the 95% band widens from plus-or-minus 1.96 to plus-or-minus 2.45. Far out, the forecast is just the mean, plus or minus the unconditional spread.

The Benchmark, and Returns

The bar, and the humbling truth

95% band : ± 1.96 ($h=1$) $\rightarrow \pm 2.45$ ($h=\infty$); must beat the naive/mean forecast

The benchmark rule

- Compare to the naive or mean forecast.
- Beating it is the minimum standard.
- In-sample fit doesn't count.

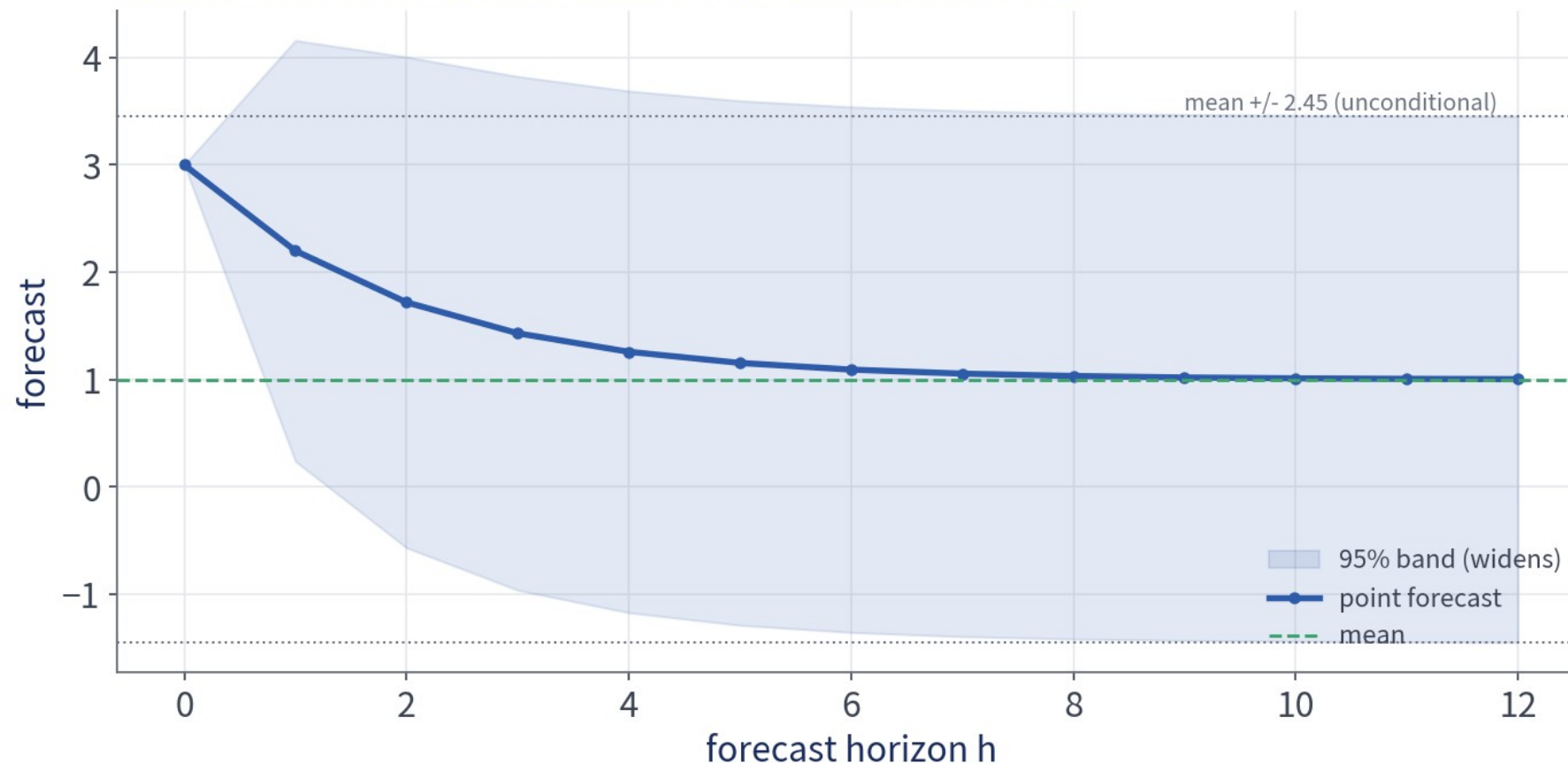
The humbling truth

- Returns carry very little ARMA structure.
- So the mean forecast is hard to beat.
- Honesty about this saves real money.

The Band Fans Out

The point forecast (blue) decays to the mean while its 95% band widens with the horizon, approaching the dotted unconditional limits of the mean plus or minus 2.45. Far out, the forecast says only 'the mean, give or take the unconditional spread' - which is to say, almost nothing

Case B: bands widen to the unconditional sd - far out, only the mean remains



Forecasting Returns, Honestly

Structure is real, but thin

Where ARMA earns its keep

- Volume, volatility, and macro series have structure.
- There, the forecast beats naive.
- The band tells you how much to trust it.

Where it humbles you

- Return levels are close to a random walk.
- The mean forecast is brutally hard to beat.
- Admitting this is the mark of a pro.

The Forecasting Checklist

Point, interval, benchmark horizon

Forecasting from a fitted ARMA has a fixed discipline. Iterate the point forecast forward, knowing it decays to the mean. Attach an honest interval that widens to the unconditional spread. Judge it out of sample against the naive benchmark - and never trust the in-sample fit alone. And ask how far ahead the forecast still says

The forecast (Case A)

- Decays 2.2, 1.72, 1.43 to the mean 1.0.
- The last value's influence fades.
- A finite information horizon.

The interval (Case B)

- Band widens from 1.96 to 2.45.
- Levels off at the unconditional spread.
- Must still beat naive to matter.

Reproduce the Forecast and the Band

Direct Claude Code to check your arithmetic

GOAL Iterate the AR(1) forecast, build the widening interval, and beat a benchmark.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that, for an AR(1)
with phi=0.6, mu=1.0, sigma^2=1, last value 3.0:
CASE A - point forecast:
    * print y_hat for h = 1..10 and confirm it decays
      to the mean.
CASE B - interval and benchmark:
    * print the error variance and 95% band for
      h = 1, 2, 3, and the limit;
    * compare an ARMA forecast to the mean forecast
      on a simulated return series (expect little gain).
Then plot the decaying forecast with its widening band.
```

Check against the slides

- forecast 2.2, 1.72, 1.43 -> 1.0.
- band +/-1.96 -> +/-2.45.
- variance limit 1.5625.
- returns barely beat the mean.

Deliverable: part3_practice.py whose printed numbers match the forecast decay and the widening band.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Where does a stationary ARMA forecast head?

Toward the long-run mean, geometrically

Q2 What does the prediction band approach?

The unconditional standard deviation

Q3 What must a forecast beat to be useful?

The naive or mean benchmark, out of sample

All three come from the forecast decay, the widening interval, and the benchmark rule.

Three Pitfalls to Avoid

Where forecasting trips people

Common mistakes

- "A long-horizon forecast is still informative."
- "Report the point forecast without a band."
- "A good in-sample fit means good forecasts."

The reality

- Far out, the forecast is just the mean.
- Always attach a widening interval.
- Only an out-of-sample win over naive counts.

Two Calculations to Remember

Part 3 applications in one screen

1

The forecast decays to the mean

From a last value of 3.0, the AR(1) forecast fell to 2.2, then 1.72, then 1.43 - each step erasing 40% of the gap - converging to the mean of 1.0. Past a horizon, the last value tells you nothing and the mean is the best guess.

2

The band widens to the unconditional spread

The error variance climbed from 1.0 toward the unconditional 1.5625, so the 95% band widened from plus-or-minus 1.96 to plus-or-minus 2.45. Far out, the forecast is only the mean plus or minus that unconditional spread.

3

Beat the benchmark, or stay humble

A forecast counts only if it beats naive out of sample - and returns, with almost no ARMA structure, rarely do. ARMA shines on volume and volatility; on return levels, honesty beats false confidence.

Week 3 complete — from AR and MA through identification to forecasting, its intervals, and its honest limits.